

Homework 2

Due: Saturday, 15th June 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

Question 1.

Prove that there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers (repetitions are allowed).

Question 2.

Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a , b , and c are real numbers.

Question 3.

Prove using the notion without loss of generality that $5x + 5y$ is an odd integer when x and y are integers of opposite parity (one is even and another is odd).

Question 4.

Prove that there is a positive integer x that equals the sum of the positive integers less than x . Is your proof constructive or nonconstructive?

Question 5.

Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?

Question 6.

Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or nonconstructive? [Hint: Do not try to evaluate these numbers!]

Question 7.

Prove or disprove that if a and b are rational numbers, then a^b is also rational.

Question 8.

Show that if a , b , and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax + b = c$.

Question 9.

Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $1/2$.

Question 10.

Prove that given a real number x there exist unique numbers n and ε such that $x = n + \varepsilon$, n is an integer, and $0 \leq \varepsilon < 1$.

Question 11.

Use forward reasoning to show that if x is a nonzero real number, then $x^2 + 1/x^2 \geq 2$. [Hint: Start with the inequality $(x - 1/x)^2 \geq 0$ which holds for all nonzero real numbers x .]

Question 12.

The harmonic mean of two real numbers x and y equals $2xy/(x + y)$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Question 13.

Formulate a conjecture about the decimal digits that appear as the final decimal digit of the fourth power of an integer. Prove your conjecture using a proof by case.

Question 14.

Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.

Question 15.

Prove that there are infinitely many solutions in positive integers x , y , and z to the equation $x^2 + y^2 = z^2$. [Hint: Let $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, where m and n are integers.]